MSG No. 123.345 $P4/mm'm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `1a`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `1b`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `1c`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `1d`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `mm'm'`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,8,9,10,13,14]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,6,7,11,12,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `mm'm'`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,8,9,10,13,14]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,6,7,11,12,15,16]`

Table 7: Wyckoff site: `2g`, site symmetry: `4m'm'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4,13,14,15,16]`
2	$[0, 0, -z]$	`[5,6,7,8,9,10,11,12]`

Table 8: Wyckoff site: 2h, site symmetry: 4m'm'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[5,6,7,8,9,10,11,12]

Table 9: Wyckoff site: 4i, site symmetry: 2m'm'.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,4,13,14]
2	$\left[\frac{1}{2}, 0, z\right]$	[2,3,15,16]
3	$\left[0, \frac{1}{2}, -z\right]$	[5,8,9,10]
4	$\left[\frac{1}{2}, 0, -z\right]$	[6,7,11,12]

Table 10: Wyckoff site: 4j, site symmetry: m.2'm'

No.	position	mapping
1	$[x, x, 0]$	[1,8,11,16]
2	$[-x, x, 0]$	[2,7,10,13]
3	$[x, -x, 0]$	[3,6,9,14]
4	$[-x, -x, 0]$	[4,5,12,15]

Table 11: Wyckoff site: 4k, site symmetry: m.2'm'

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	[1,8,11,16]
2	$\left[-x, x, \frac{1}{2}\right]$	[2,7,10,13]
3	$\left[x, -x, \frac{1}{2}\right]$	[3,6,9,14]
4	$\left[-x, -x, \frac{1}{2}\right]$	[4,5,12,15]

Table 12: Wyckoff site: 4l, site symmetry: m2'm'.

No.	position	mapping
1	$[x, 0, 0]$	[1,8,9,14]
2	$[0, x, 0]$	[2,7,11,16]
3	$[0, -x, 0]$	[3,6,12,15]
4	$[-x, 0, 0]$	[4,5,10,13]

Table 13: Wyckoff site: 4m, site symmetry: m2'm'.

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,8,9,14]
2	$\left[0, x, \frac{1}{2}\right]$	[2,7,11,16]
3	$\left[0, -x, \frac{1}{2}\right]$	[3,6,12,15]

continued ...

Table 13

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[4,5,10,13]`

Table 14: Wyckoff site: `4n`, site symmetry: `m2'm'`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,8,9,14]`
2	$\left[\frac{1}{2},\, x,\, 0\right]$	`[2,7,11,16]`
3	$\left[\frac{1}{2},\, -x,\, 0\right]$	`[3,6,12,15]`
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[4,5,10,13]`

Table 15: Wyckoff site: `4o`, site symmetry: `m2'm'`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[1,8,9,14]`
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[2,7,11,16]`
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[3,6,12,15]`
4	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[4,5,10,13]`

Table 16: Wyckoff site: `8p`, site symmetry: `m..`.

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,8]`
2	$[-y,\, x,\, 0]$	`[2,7]`
3	$[y,\, -x,\, 0]$	`[3,6]`
4	$[-x,\, -y,\, 0]$	`[4,5]`
5	$[x,\, -y,\, 0]$	`[9,14]`
6	$[-x,\, y,\, 0]$	`[10,13]`
7	$[y,\, x,\, 0]$	`[11,16]`
8	$[-y,\, -x,\, 0]$	`[12,15]`

Table 17: Wyckoff site: `8q`, site symmetry: `m..`.

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,8]`
2	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[2,7]`
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[3,6]`
4	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[4,5]`
5	$\left[x,\, -y,\, \frac{1}{2}\right]$	`[9,14]`
6	$\left[-x,\, y,\, \frac{1}{2}\right]$	`[10,13]`
7	$\left[y,\, x,\, \frac{1}{2}\right]$	`[11,16]`
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	`[12,15]`

Table 18: Wyckoff site: `8r`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16]`
2	$[-x,\ x,\ z]$	`[2,13]`
3	$[x,\ -x,\ z]$	`[3,14]`
4	$[-x,\ -x,\ z]$	`[4,15]`
5	$[-x,\ -x,\ -z]$	`[5,12]`
6	$[x,\ -x,\ -z]$	`[6,9]`
7	$[-x,\ x,\ -z]$	`[7,10]`
8	$[x,\ x,\ -z]$	`[8,11]`

Table 19: Wyckoff site: `8s`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,14]`
2	$[0,\ x,\ z]$	`[2,16]`
3	$[0,\ -x,\ z]$	`[3,15]`
4	$[-x,\ 0,\ z]$	`[4,13]`
5	$[-x,\ 0,\ -z]$	`[5,10]`
6	$[0,\ -x,\ -z]$	`[6,12]`
7	$[0,\ x,\ -z]$	`[7,11]`
8	$[x,\ 0,\ -z]$	`[8,9]`

Table 20: Wyckoff site: `8t`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\ x,\ z\right]$	`[2,16]`
3	$\left[\frac{1}{2},\ -x,\ z\right]$	`[3,15]`
4	$\left[-x,\ \frac{1}{2},\ z\right]$	`[4,13]`
5	$\left[-x,\ \frac{1}{2},\ -z\right]$	`[5,10]`
6	$\left[\frac{1}{2},\ -x,\ -z\right]$	`[6,12]`
7	$\left[\frac{1}{2},\ x,\ -z\right]$	`[7,11]`
8	$\left[x,\ \frac{1}{2},\ -z\right]$	`[8,9]`

Table 21: Wyckoff site: `16u`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x,\ z]$	`[2]`
3	$[y,\ -x,\ z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$[y,\ -x,\ -z]$	`[6]`
7	$[-y,\ x,\ -z]$	`[7]`

continued ...

Table 21

No.	position	mapping
8	$[x, y, -z]$	`[8]`
9	$[x, -y, -z]$	`[9]`
10	$[-x, y, -z]$	`[10]`
11	$[y, x, -z]$	`[11]`
12	$[-y, -x, -z]$	`[12]`
13	$[-x, y, z]$	`[13]`
14	$[x, -y, z]$	`[14]`
15	$[-y, -x, z]$	`[15]`
16	$[y, x, z]$	`[16]`